

Phy 211

Lecture 8 Notes

By

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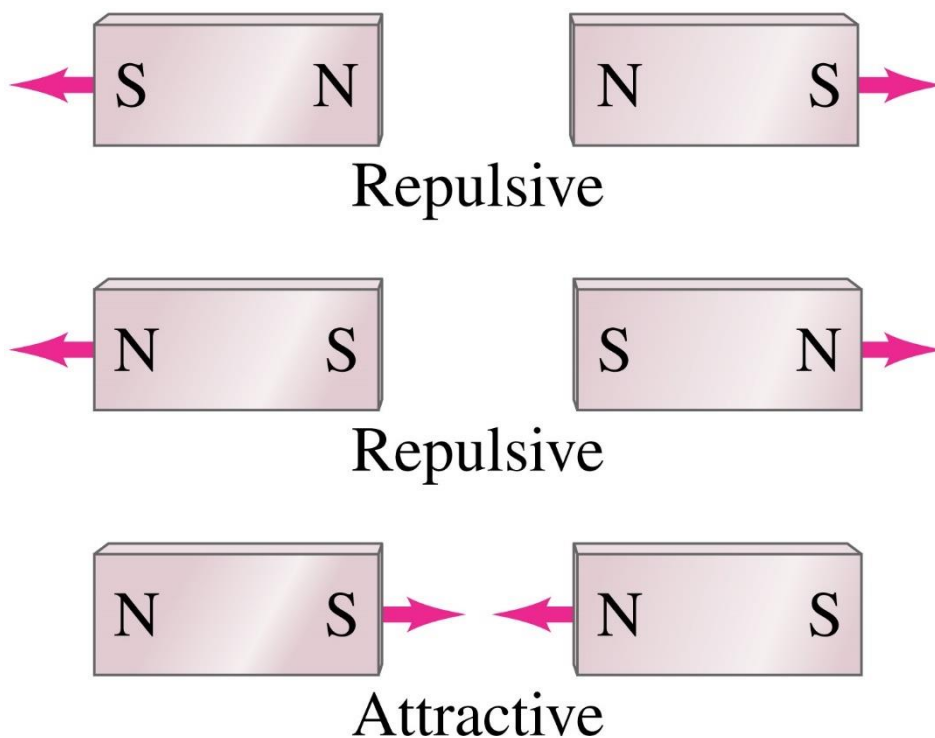
Fall 2024

Chapter 7

Magnetic Fields

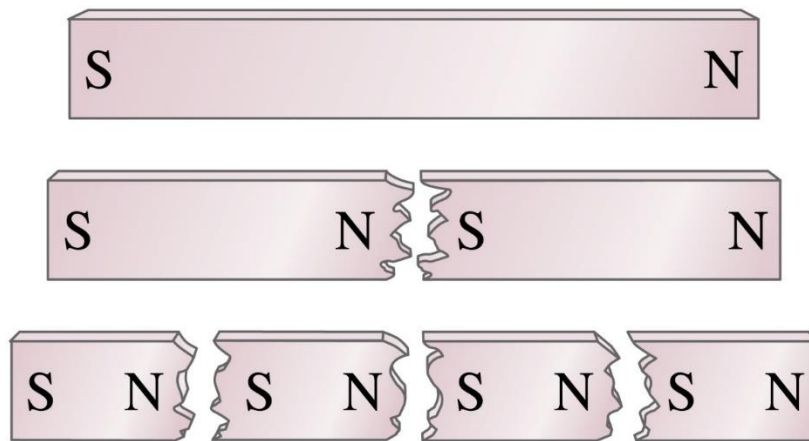
➤ Magnetic poles:

- Every magnet, regardless of its shape, has two poles called north and south poles.
- Poles exert forces on one another similar to the way electric charges exert forces on each other:
 - Like poles repel each other (N-N or S-S)
 - Unlike poles attract each other (N-S)



- The poles received their names due to the way a magnet behaves in the Earth's magnetic field
- If a bar magnet is suspended so that it can move freely, it will rotate so that the magnetic north pole points toward the Earth's north geographic pole.

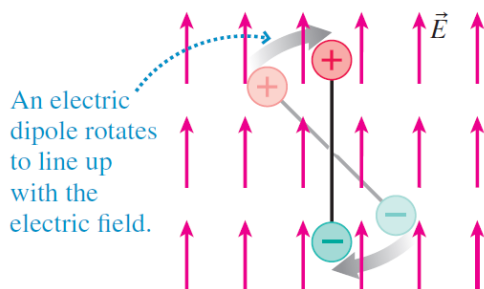
- The force between two poles varies as the inverse square of the distance between them.
- A single magnetic pole has never been isolated (**No existence of magnetic monopole**).
- No matter how many times a permanent magnetic is cut in two, each piece always has a north and south pole.



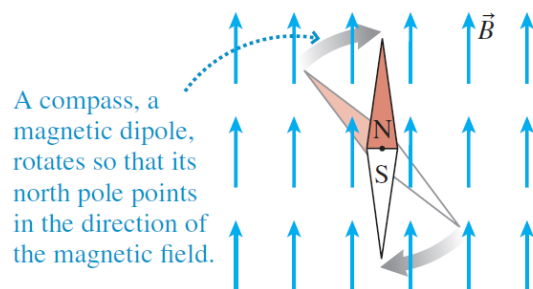
➤ Magnetic fields:

- Reminder: an electric field surrounds any electric charge.
- The region of space surrounding any moving electric charge also contains a magnetic field.
- **Every magnet sets up a *magnetic* field in the space around it.**
- The direction of the magnetic field $\vec{B}$ at any location is the direction in which a north pole of a compass needle points at that location.

(a) Electric field



(b) Magnetic field



- As with the electric field, we can represent the magnetic field by means of drawings with magnetic field lines.
- The compass can be used to trace the field lines.
- The lines outside the magnet point from the **North pole** to the **South pole**

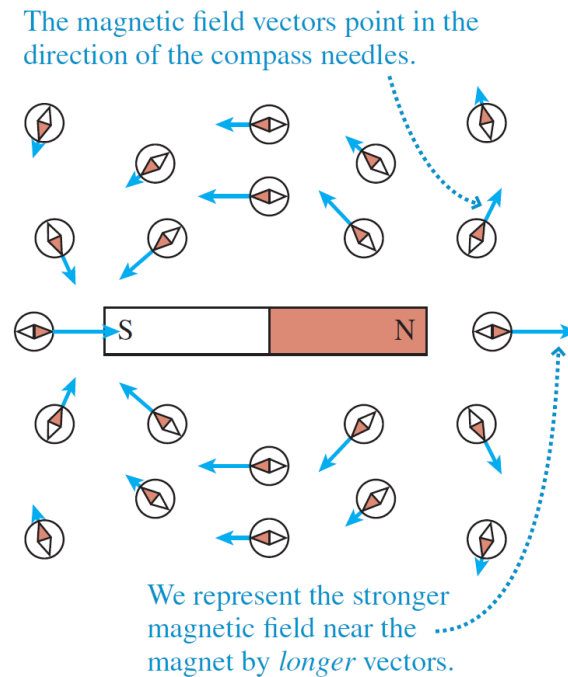
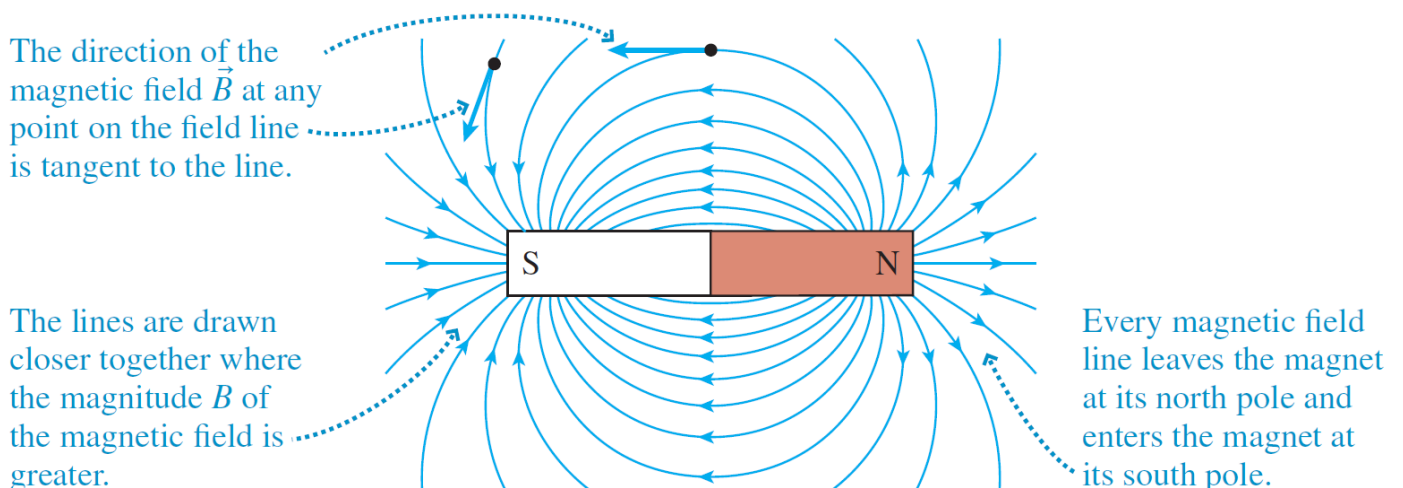
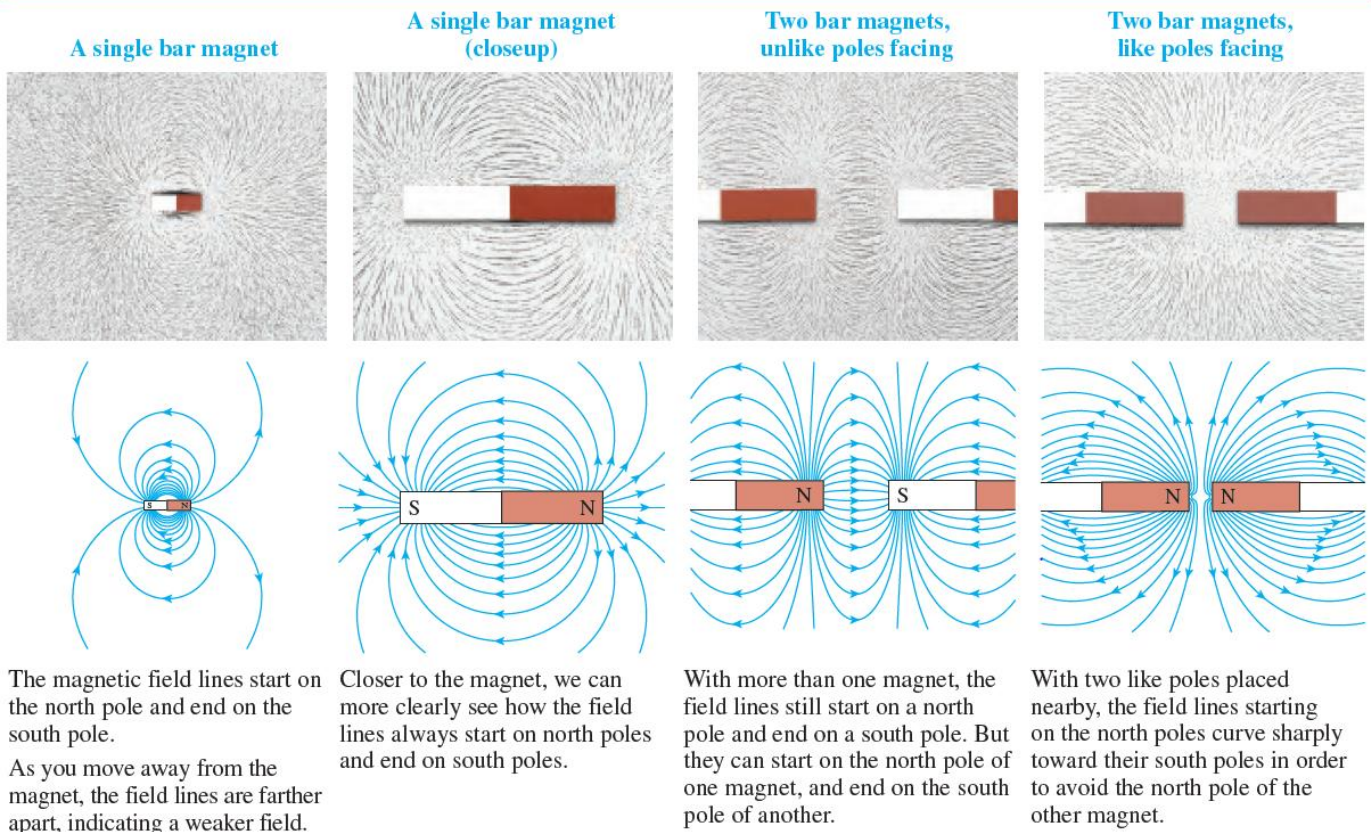


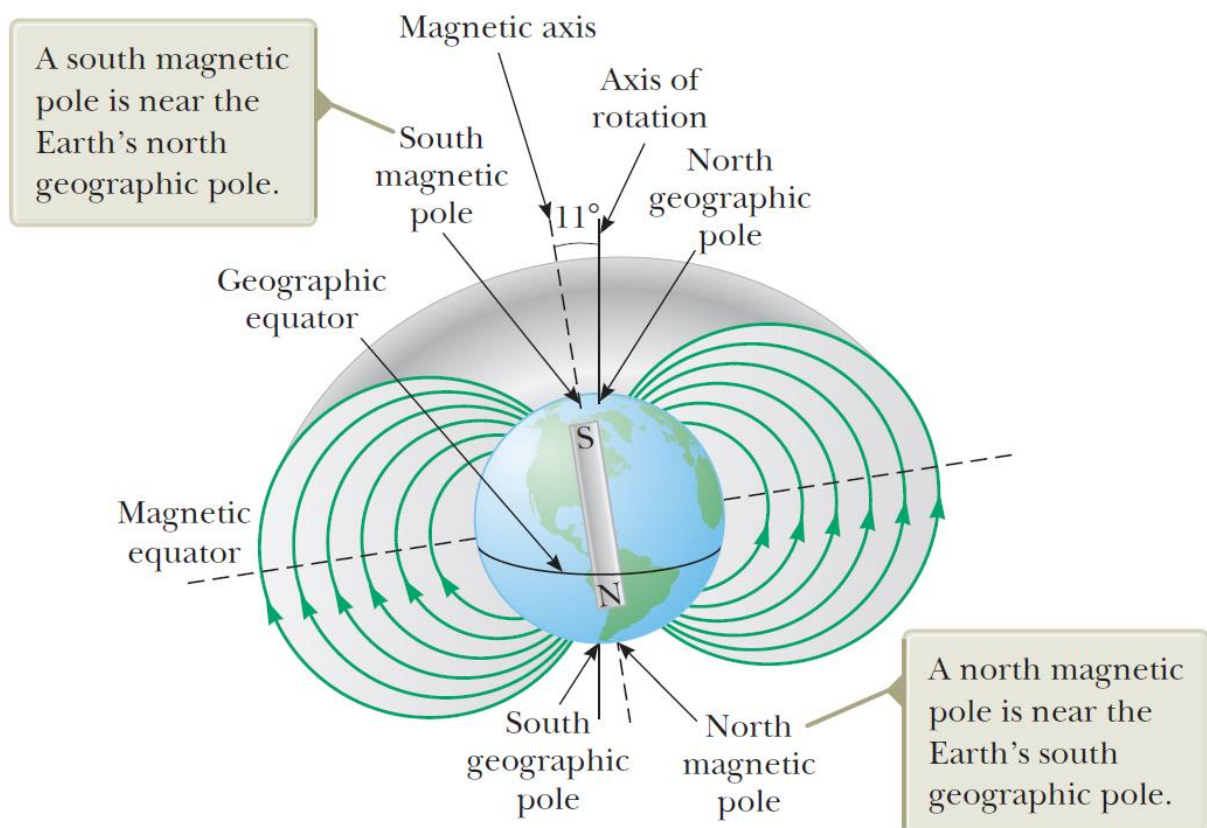
FIGURE Drawing the magnetic field lines of a bar magnet.



An atlas of magnetic fields produced by magnets



• The Earth's Magnetic Field:



➤ Force on a Charge Moving in a Magnetic Field

- The magnitude F_B of the magnetic force exerted on the particle is proportional to the charge q , magnetic field B and speed of the particle v .
- When a charged particle moves parallel to the magnetic field vector, the magnetic force acting on the particle is zero
- When the particle's velocity vector makes any angle $\theta \neq 0$ with the field, the force acts in a direction perpendicular to both the velocity and the field.
- The magnetic force exerted on a positive charge is in the direction opposite the direction of the magnetic force exerted on a negative charge moving in the same direction
- The magnitude of the magnetic force is proportional to $\sin \theta$, where θ is the angle the particle's velocity makes with the direction of the magnetic field.

The force on a charged particle moving in a magnetic field

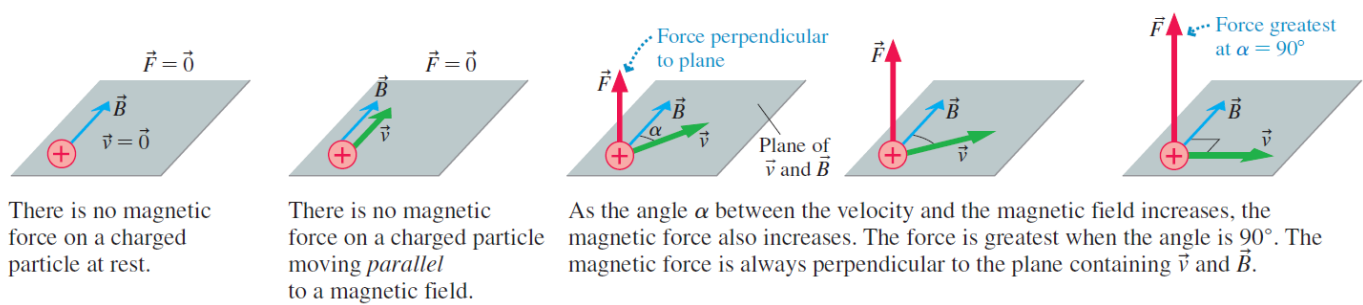
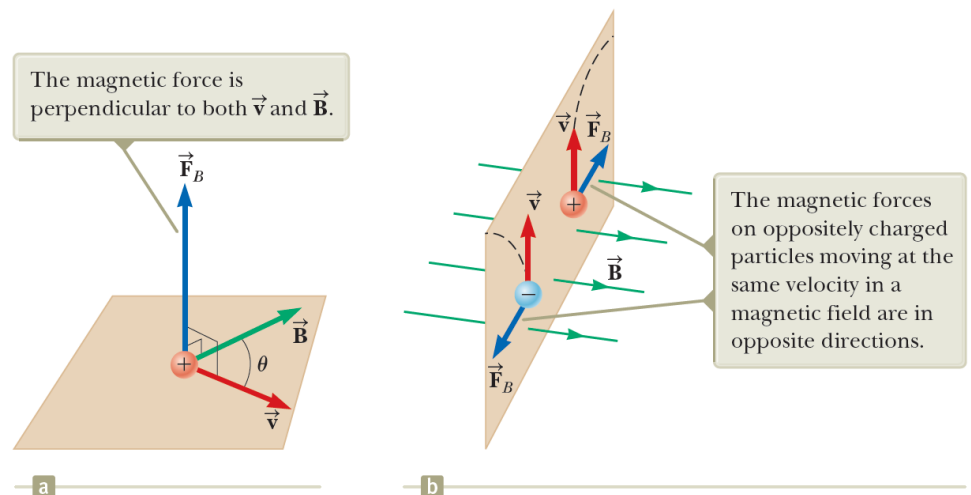


Figure (a) The direction of the magnetic force $\vec{F}_B$ acting on a charged particle moving with a velocity $\vec{v}$ in the presence of a magnetic field $\vec{B}$. (b) Magnetic forces on positive and negative charges. The dashed lines show the paths of the particles.



- The properties can be summarized in a vector equation:

$$\vec{F}_B = q\vec{v} \times \vec{B}$$

$\vec{F}_B$: magnetic force (Unit: N)

q : the charge (Unit: C)

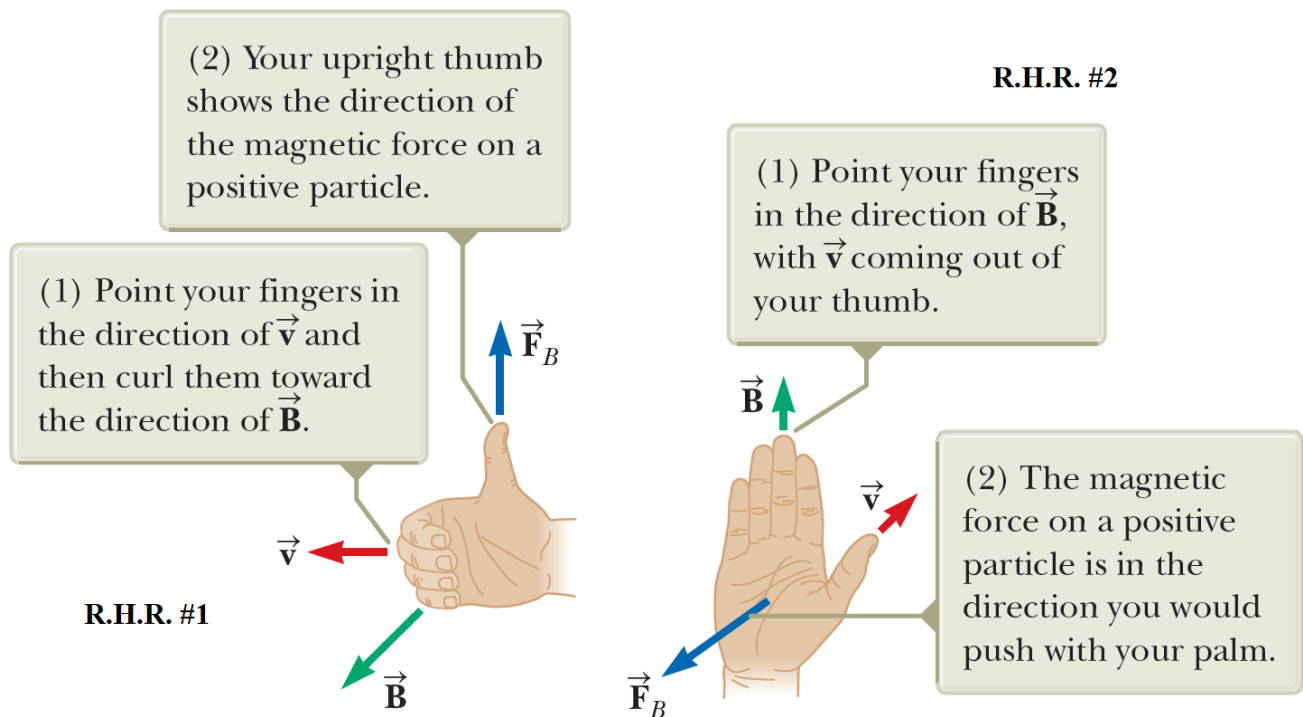
$\vec{v}$: the velocity of the moving charge (Unit: m/s)

$\vec{B}$: the magnetic field (Unit: T)

- The magnitude of the magnetic force on a charged particle is

$$F_B = |q|vB \sin \theta$$

- θ is the smaller angle between v and B
- F_B is minimum ($(F_B)_{min} = \text{zero}$) when the field and velocity are parallel or antiparallel ($\theta = 0 \text{ or } 180^\circ$)
- F_B is a maximum ($(F_B)_{max} = |q|vB$) when the field and velocity are perpendicular ($\theta = 90^\circ$)
- The direction of the magnetic force can be determined by **Right-Hand rule 1 or 2**.



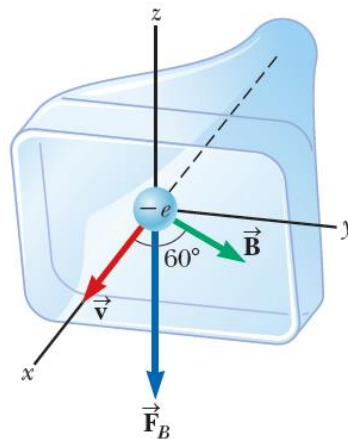
- The SI unit of magnetic field is the newton per coulomb-meter per second, which is called the tesla (T):

$$1 \text{ T} = 1 \frac{\text{Wb}}{\text{m}^2} = 1 \frac{\text{N}}{\text{C} \cdot \frac{\text{m}}{\text{s}}} = 1 \frac{\text{N}}{\text{A} \cdot \text{m}}$$

- A non-SI magnetic-field unit in common use, called the gauss (G), is related to the tesla through the conversion

$$1 \text{ T} = 10^4 \text{ Gauss}$$

- Example 1:** An electron in an old-style television picture tube moves toward the front of the tube with a speed of $8.0 \times 10^6 \text{ m/s}$ along the x axis (Figure below). Surrounding the neck of the tube are coils of wire that create a magnetic field of magnitude 0.025 T , directed at an angle of 60° to the x axis and lying in the xy plane. Calculate the magnetic force on the electron.



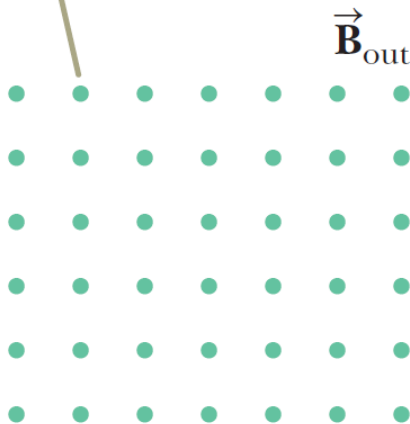
Solution

$$F_B = |q|vB \sin \theta = (1.6 \times 10^{-19}) \times (8.0 \times 10^6) \times (0.025) \times \sin 60$$

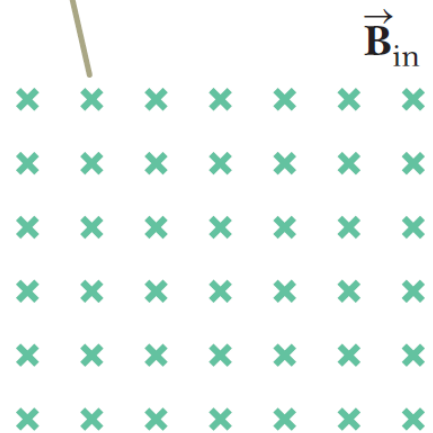
$$F_B = 2.8 \times 10^{-14} \text{ N}$$

- **Notation notes:**

Magnetic field lines coming out of the paper are indicated by dots, representing the tips of arrows coming outward.



Magnetic field lines going into the paper are indicated by crosses, representing the feathers of arrows going inward.

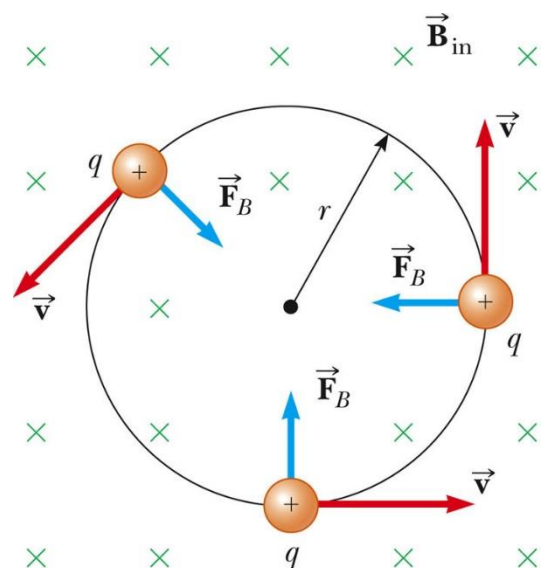


➤ **Motion of a charged particle in a uniform magnetic field:**

- Consider a particle moving in an external magnetic field with its velocity perpendicular to the field.
- The force is always directed toward the center of the circular path.
- The magnetic force causes a centripetal acceleration, changing the direction of the velocity of the particle.

$$F_B = |q|vB = \frac{mv^2}{r}$$

$$r = \frac{mv}{|q|B}$$



- That is, the radius of the path is proportional to the linear momentum " mv " of the particle and inversely proportional to the magnitude of the charge on the particle and to the magnitude of the magnetic field.
- The angular speed of the particle is

$$\omega = \frac{v}{r} \left(\text{Unit: } \frac{\text{rad}}{\text{s}} \right)$$

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- **Example 2:** A proton is moving in a circular orbit of radius 14 cm in a uniform 0.35 T magnetic field perpendicular to the velocity of the proton. Find the speed of the proton. ($m_p = 1.67 \times 10^{-27}$ kg)

Solution

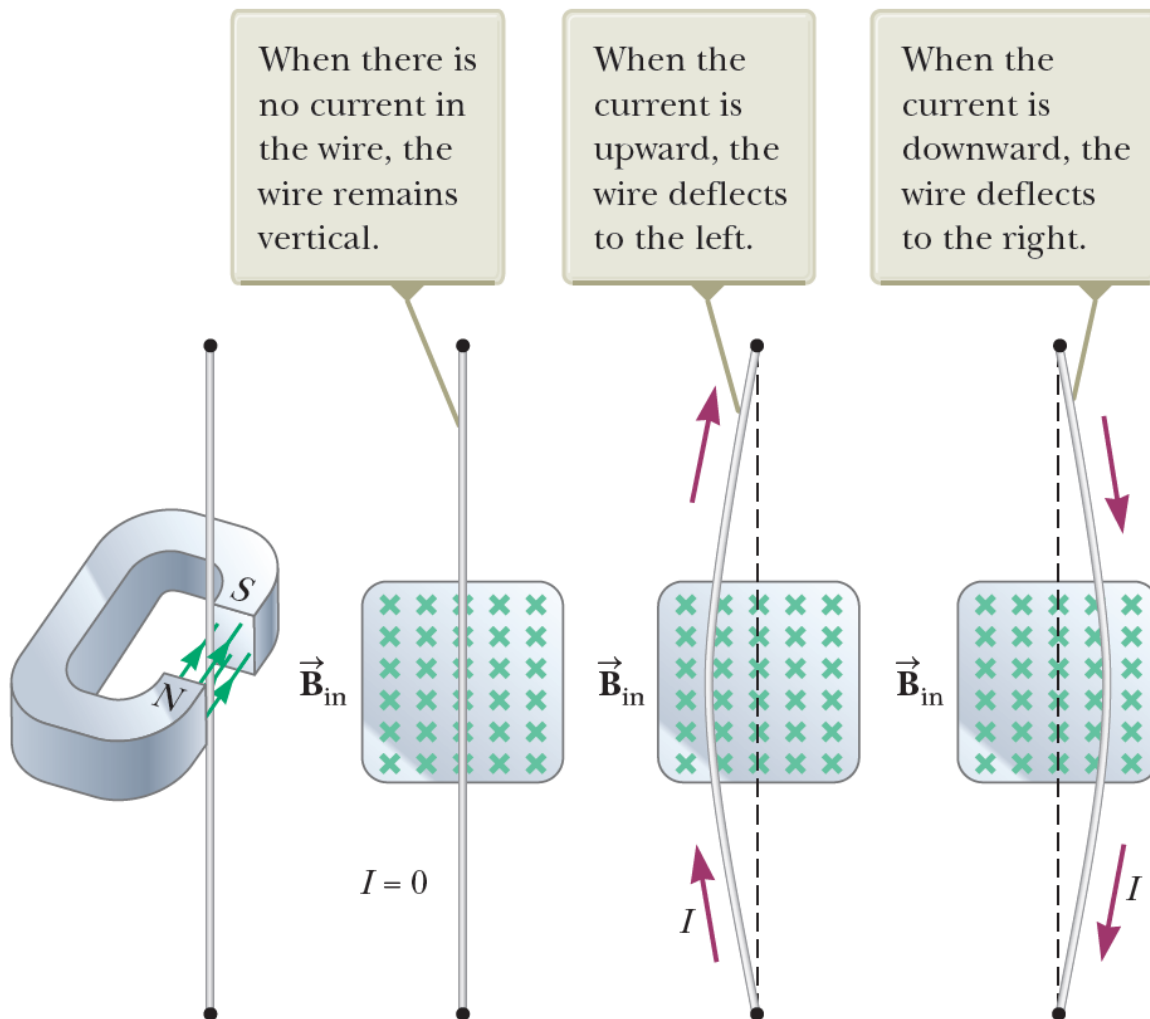
$$r = \frac{mv}{|q|B} \rightarrow v = \frac{|q|rB}{m}$$

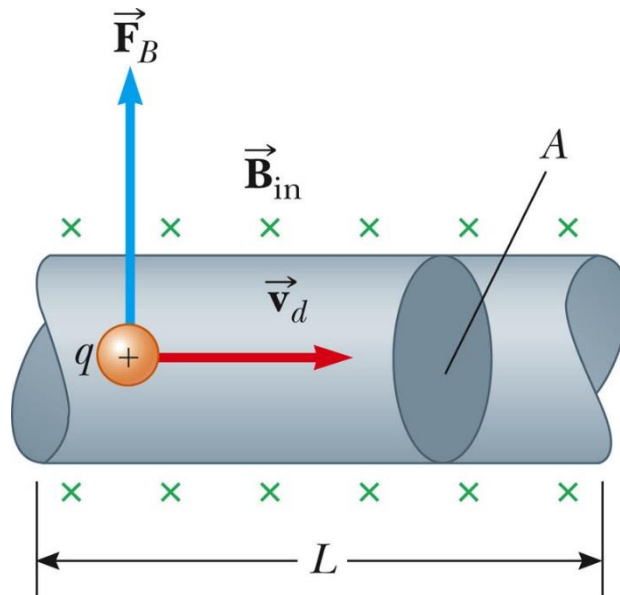
$$v = \frac{|q|rB}{m} = \frac{(1.6 \times 10^{-16}) \times (0.14) \times (0.35)}{1.67 \times 10^{-27}}$$

$$v = 4.7 \times 10^6 \text{ m/s}$$

➤ Magnetic force acting on a current-carrying conductor:

- The current is a collection of many charged particles in motion; hence, the resultant force exerted by the field on the wire is the vector sum of the individual forces exerted on all the charged particles making up the current.





- To find the total force acting on the wire, we multiply the force $q\vec{v} \times \vec{B}$ exerted on one charge by the number of charges in the segment.
- Because the volume of the segment is AL , the number of charges in the segment is nAL , where n is the number of mobile charge carriers per unit volume.

$$\vec{F}_B = (q\vec{v} \times \vec{B})nAL$$

$$\& I = nqA\vec{v}$$

$$\boxed{\therefore \vec{F}_B = I\vec{L} \times \vec{B}}$$

- I is the current.
- $\vec{L}$ is a vector that points in the direction of the current. Its magnitude is the length L of the segment.
- $\vec{B}$ is the magnetic field.

$$\boxed{F_B = BIL \sin \theta}$$

- θ is the angle between $\vec{L}$ & $\vec{B}$. (i.e. I & $\vec{B}$)